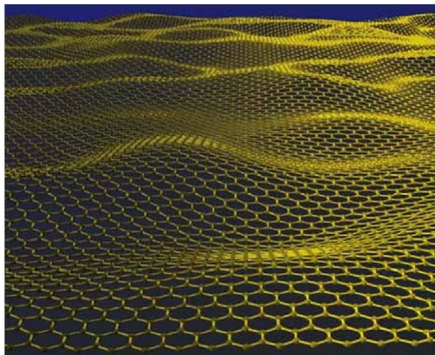


CAN GRAPHENE ELECTRONICS TAKE OVER?

From graphene, scientists expect to make nano-electronic devices that work on diffraction of electrons rather than diffusion of electrons. These devices are predicted to be substantially faster, thinner and more efficient than current silicon-based devices



Graphene structure

■ DR S.S. VERMA

Carbon—the basis of all known life on the earth—has surprised the scientific community once again with its exotic properties. Researchers have found that a form of carbon called graphene makes experiments possible that give new twists to the phenomena in quantum physics.

Owing to its unusual electronic spectrum, graphene has led to the emergence of a new paradigm of relativistic condensed-matter physics, where quantum relativistic phenomena, some of which are unobservable in

high-energy physics, can now be mimicked and tested in table-top experiments. More generally, it offers new inroads into low-dimensional physics.

Also, a vast variety of practical applications now appear possible including the creation of new materials and the manufacture of innovative electronics. Graphene electronic devices are predicted to be substantially faster, thinner and more efficient than current silicon-based devices.

Graphene as a material

Graphene as a material is completely new—not only the thinnest ever but

also the strongest. It is a one-atom thick planar sheet of sp²-bonded carbon atoms, which makes it the thinnest material ever discovered. Carbon atoms in the sheet are densely packed in a two-dimensional (2D) honeycomb crystal lattice. The carbon-carbon bond length in graphene is about 0.142 nm.

Graphene is the basic structural element of some carbon allotropes including graphite, charcoal, carbon nanotubes and fullerenes. It can be wrapped up into 0D fullerenes, rolled into 1D nanotubes or stacked into 3D graphite.

Graphene can also be considered as an infinitely large aromatic molecule—the limiting case of the family of flat polycyclic aromatic hydrocarbons called graphenes. At an atomic scale, it looks a bit like a chicken wire made of carbon atoms and their bonds.

Graphene is highly conductive—conducting both heat and electricity better than any other material, including copper—and stronger than diamond. It is almost completely transparent, yet so dense that not even helium (the smallest gas atom) can pass through it. With graphene, physicists can now study a new class of two-dimensional materials with unique properties.

Two Russian-born scientists based at the University of Manchester in the UK shared the 2010 Nobel Prize for Physics for their groundbreaking work on graphene. They have shown that carbon in such a flat form has exceptional properties that originate from the remarkable world of quantum physics.